SE LAB 7

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SECTION: F

1. Running the buggy application

```
PS C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7> python vending_machine.py
==== Running parse_coins ====
Enter coin values as CSV (e.g. 5,10,25): 5,10,25
Coins: [5, 10, 25]

==== Running slot_position ====
Enter a slot code (e.g. B7): B7
Row and Column: (1, 6)

==== Running change_due ====
Enter paid/price (e.g. 200/150): 200/150
Change Due: 10 five-coins
```

2. Errors after running the test file

```
+ Exception Group Traceback (most recent call last):
|   File "C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7\test_vending_machine.py", line 19, in test_change_due_no_crash
|     def test_change_due_no_crash(s):
|                    ^^^
| ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7\test_vending_machine.py", line 21, in test_change_due_no_crash
  |     change_due(s)
  |     ^^^^^^^^^^^^^
  |   File "C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7\vending_machine.py", line 11, in change_due
  |     change = int(paid) - int(price)
  |              ~~~^^^^^^
  | ValueError: invalid literal for int() with base 10: ''
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7\test_vending_machine.py", line 23, in test_change_due_no_crash
  |     assert False, f"change_due crashed with: {e!r}"
  | AssertionError: change_due crashed with: ValueError("invalid literal for int() with base 10: ''")
  | assert False
  | Failing test case: test_change_due_no_crash(
  |     s='/',
  | )
```

```
+---------------- 2 ----------------
| Traceback (most recent call last):
|   File "C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7\test_vending_machine.py", line 21, in test_change_due_no_crash
|     change_due(s)
|     ~~~~~~~~~~^^^
|   File "C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7\vending_machine.py", line 10, in change_due
|     paid, price = payment.split('/')
|     ^^^^^^^^^^^
| ValueError: not enough values to unpack (expected 2, got 1)
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7\test_vending_machine.py", line 23, in test_change_due_no_crash
|     assert False, f"change_due crashed with: {e!r}"
| AssertionError: change_due crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
| assert False
| Failing test case: test_change_due_no_crash(
|     s='',  # or any other generated value
| )
+---------------- 3 ----------------
```

```
+---------------- 3 ----------------
| Traceback (most recent call last):
|   File "C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7\test_vending_machine.py", line 21, in test_change_due_no_crash
|     change_due(s)
|     ~~~~~~~~~~^^^
|   File "C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7\vending_machine.py", line 10, in change_due
|     paid, price = payment.split('/')
|                   ^^^^^^^^^^^^^
| AttributeError: 'NoneType' object has no attribute 'split'
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7\test_vending_machine.py", line 23, in test_change_due_no_crash
|     assert False, f"change_due crashed with: {e!r}"
| AssertionError: change_due crashed with: AttributeError("'NoneType' object has no attribute 'split'")
| assert False
| Failing test case: test_change_due_no_crash(
|     s=None,  # or any other generated value
| )
+-----------------------------------
```

```
=========================================================================================== short test summary info ============================
FAILED test_vending_machine.py::test_parse_coins_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_vending_machine.py::test_slot_position_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_vending_machine.py::test_change_due_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================================================================================ 3 failed in 8.36s =============================
```

3. After fixing vending_machine.py

```
PS C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7> python vending_machine.py
==== Running parse_coins ====
Enter coin values as CSV (e.g. 5,10,25): abc
Coins: []

==== Running slot_position ====
Enter a slot code (e.g. B7): Z
Row and Column: Invalid slot code

==== Running change_due ====
Enter paid/price (e.g. 200/150): 50/100
Change Due: Invalid payment
```

4. Rerunning the tests

```
PS C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7> pytest test_vending_machine.py
============================================================================= test session starts ==============================================================================
platform win32 -- Python 3.14.6, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Paridhi\Desktop\colej\SE\se6\L6_L7_30_vending_machine\lab7
plugins: hypothesis-6.170.15
collected 3 items

test_vending_machine.py ...                                                                                                                                          [100%]

============================================================================== 3 passed in 1.01s ===============================================================================
```